P1 2025

$128 = 2^7$

⇒ 16 biți

(1.) S.M.

Sign

$x = -\frac{55}{64} = -\frac{110}{128} = 1\,1101110 \cdot 2^{-7}$ ⇒ count 7

$y = -\frac{103}{128} = -\frac{103}{128} = 1\,110\,0111 \cdot 2^{-7}$

⟶ mutăm virgula la stânga cu 7 unități la final

$P = x \cdot y = (-110) \cdot (-103) \cdot 2^{-14} = 0\,010\,1100\;0100\;0010 \cdot 2^{-14}$

$= 0.10\,1100\;0100\;0010$

$= 0.69\,15283203$ ± eroare

```
  103·
  110
 ----
  000
 103
103
-----
11330
```

A	Q	M	COUNT	ACS
00000000		1110 0111	000	C_0
	1110 1110			C_1
00000000	0111 0111		001	C_3
⊕ 1100111				
01100111				C_2
0011001 1	1011 1011		010	C_3
⊕ 1100111				
10011010				C_2
0100110 1	0101 1101		011	C_3
⊕ 1100111				
10110100				C_2
01011010	0010 1110		100	C_3
0010110 1	0001 0111		101	C_3
⊕ 1100111				

10010100				C_2
0100101 0	0000	1011	110	C_3
⊕ 1100111				
10110001				C_2
01011000	1000	0101	111 count 7	C_3
00101100	0100	0010		C_4
0010 1100				C_5
	0100	0010		C_6

EXOR

$$\Rightarrow P = \underset{8192}{0010}\ \underset{4096\ 2048}{1100}\ \underset{1024\ 512}{0100}\ \underset{256\ 128}{0010}\ \underset{64\ 32}{}\ \underset{16\ 8\ 4\ 2\ 1}{}\cdot 2^{-14} =$$

$$= \frac{2+64+1024+2048+8192}{2^{14}} = \frac{11330}{2^{14}} \approx 0{,}69$$

ca sus

2. $x = -\frac{55}{64} = -\frac{110}{128} = \overset{\text{Sign}}{1}\,1101110_{(SM)}\cdot 2^{-7} = 1\,0010010_{(C_2)}\cdot 2^{-7}$

$y = -\frac{103}{128} = -\frac{103}{128} = 1\,110\,0111_{(SM)}\cdot 2^{-7} = 1\,001\,1001_{(C_2)}\cdot 2^{-7}$

$P = x\cdot y = (-110)\cdot(-103)\cdot 2^{-14} = 0\,010\,1100\ \ 0100\ \ 0010\cdot 2^{-14}$

$= 0.10\,1100\ \ 0100\ \ 0010$

$= 0.69\,15283203 \pm$ eroare

F	A	Q	M	COUNT	ACS
0	0000 0000		1001 1001	000	C_0
		1001 0010			C_1
	F 0000 0000	0100 1001	AMB	001 ≠7	C_3
	⊕1001 1001				
1	1001 1001				C_2
	1100 1100	1010 0100		010 !=7	C_3
	1110 0110	0101 0010		011	C_3
	1111 0011	0010 1001		100	C_3
	⊕1001 1001				
1	1 1000 1100 ignorat				C_2
	1100 0110	0001 0100		101	C_3
	1110 0011	0000 1010		110	C_3
	1111 0001	1000 0101	==1 ⇒ corecție	111 ==7	C_3
	⊖1001 1001				C_2, C_4
	0101 1000				
	nr. întreg				
	0010 1100	0100 0010			
	0010 1100				C_5
		0100 0010			C_6

aceeași verificare

A	Q	M	COUNT	ACS
0000 0000		1001 1001	000	C_0
	1001 0010			C_1
0000 0000	0100 10010		001	C_4
⊖				
1001 1001				
0110 0111				C_2, C_3
00ii 00ii	1010 01001		010	C_4
1001 1001				
1100 1100				C_2
⊕				
1110 0110	0101 00100		011	C_4
1111 0011	0010 10010		100	C_4
⊖				
1001 1001				
0101 1010				C_2, C_3
00i0 1101	0001 01001		101	C_4
⊕				
1001 1001				
1100 0110				C_2
1110 0011	0000 10100		110	C_4
1111 0001	1000 01010		111	C_4
⊖				
1001 1001				
0101 1000				C_2, C_3
nr. întregi				
0010 1100	0100 00101			C_4
0010 1100				C_5
	0100 0010			C_6